Exam Assignment
02429 Analysis of correlated data: Mixed linear models
DTU Compute, Autumn 2025

November 20, 2025

To be uploaded to dtulearn on **Friday the 5th before 8 am.** Individual assignment (40% written exam assignment and 60% individual oral exam).

The report is limited to a total **maximum of 20 pages** for the main text, do include as many appendices as necessary. **In addition to the 20 pages**, include an appendix with the ranova, anova Type III and summary output for each exercise, and also include the R-code at the end of the report.

Substance is more important than style; though, keep a **concise presentation**, following **comments noted in the midterm assignment**.

Exercise I. Observational data. Radon. Multilevel linear model

Radon is a carcinogen—a naturally occurring radioactive gas whose decay products are also radioactive—known to cause lung cancer in high concentrations and estimated to cause several thousand lung cancer deaths. The distribution of radon levels in U.S. homes varies greatly. In order to identify the areas with high radon exposures, the Environmental Protection Agency coordinated radon measurements in a **random sample** of more than 80,000 houses throughout the country.

To simplify the problem, **the goal** in analyzing these data was to estimate the distribution of radon levels in each of the counties in the state of Minnesota, so that homeowners could make decisions about measuring or remediating the radon in their houses based on the best available knowledge of local conditions. For the purpose of this analysis, the data were structured hierarchically: houses within counties.

In performing the analysis, we had an important predictor—the floor on which the measurement was taken, either basement or first floor; radon comes from underground and can enter more easily when a house is built into the ground. We also had an important county-level predictor—a measurement of soil uranium that was available at the county level.

Fit a mixed model where y_i is the logarithm of the radon measurement in house i, x is the floor of the measurement (that is, 0 for basement and 1 for first floor), and u is the uranium measurement at the county level. The hierarchical model allows us to fit a regression model to the individual measurements while accounting for systematic unexplained variation among the counties. The inference is aimed at any county of the state not only at those in the sample.

The datafile and code for producing Figure 1 are in the files `Radon_MN.csv, Radon_data.R`

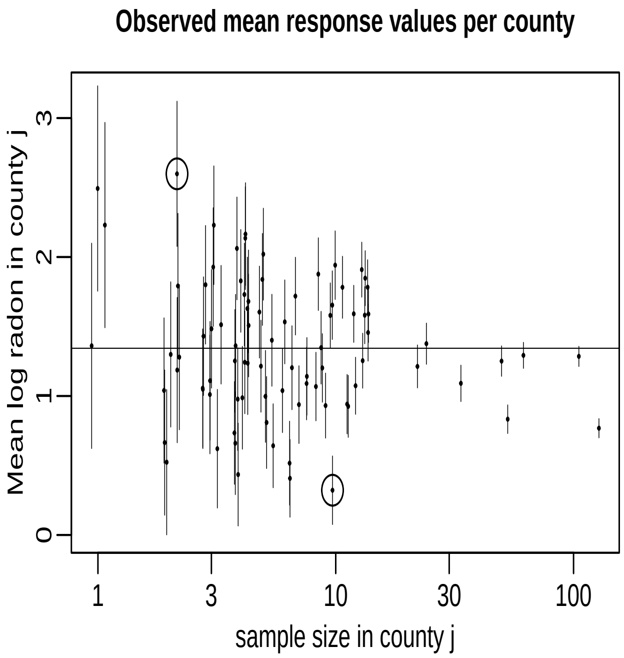

Figure 1: Observed mean response values per county with empirical confidence intervals. The sample size corresponds to the number of houses sampled per county (x-axis values have been jittered for displaying overlapping values, e.g. there are three counties with sample size 1.

I. Exploratory analysis.

Do a graphical and numerical exploratory analysis. What do you see? Figure 1 is part of the exploratory analysis.

II. Modelling.

- Start with a model and build up or down a potential final model.
- Present diagnostic plots/statistics of the model
- Present a table comparing the different models you considered in the modelling process.
- Give a mathematical expression of the starting and final model.

III Results.

- Present a plot analogous to the one in Figure 1, with the estimated response means per county.
- Present a plot of the county level of uranium in the horizontal axis, and the estimated regression intercepts with CI, in the vertical axis. Additionally, one with the estimated regression slopes with CI.
- Present any additional results you may consider relevant.
- Predictions with confidence intervals. What would be the predicted radon level for a new house in one of the observed counties?, and what for a new house in a non-observed county?, What would be the predicted mean value for an observed county? and what for a non-observed county? Give an example for specific houses/counties.

III. Conclusion.

State your main results and conclusions from your analysis.

Exercise II. Experimental data. Repeated measures/Longitudinal data.

In biological investigations the growth of an animal or part of an animal is often the subject of study; the following table shows measurements for 27 rats kept in separate cages. The rats were divided at random into 3 groups containing 10, 7, and 10 rats respectively, (the second group contains fewer rats than the other two, due to an accident at the beginning of the experiment). The first group were kept as a control, the second group had thyroxin, and the third group thiouracil added to their drinking water.

For each rat, its weight was measured before any treatment (`y0`), and after treatment, its weight gain at weeks 1, 2, 3 and 4 (`y1, y2, y3, y4`). The initial weight (`y0`) should be considered as a baseline and hence considered as an adjusting covariate in the model.

The response for each experimental unit is a sequence of measurements on a continuous scale. The mean response depends on both the experimental treatment and the time at which the response is measured. The **primary objective** is to make inferences about the effects of the experimental treatments on the mean response profile.

INITIAL WEIGHT AND WEEKLY GAINS IN WEIGHT FOR 27 RATS

Group 1. Control						Group 2. Thyroxin						Group 3. Thiouracil					
Rat	y_0	y_1	y_2	y_3	y_4	Rat	y_0	y_1	y_2	y_3	y_4	Rat	y_0	y_1	y_2	y_3	y_4
1	57	29	28	25	33	11	59	26	36	35	35	18	61	25	23	11	9
2	60	33	30	23	31	12	54	17	19	20	28	19	59	21	21	10	11
3	52	25	34	33	41	13	56	19	33	43	38	20	53	26	21	6	27
4	49	18	33	29	35	14	59	26	31	32	29	21	59	29	12	11	11
5	56	25	23	17	30	15	57	15	25	23	24	22	51	24	26	22	17
6	46	24	32	29	22	16	52	21	24	19	24	23	51	24	17	8	19
7	51	20	23	16	31	17	52	18	35	33	33	24	56	22	17	8	5
8	63	28	21	18	24							25	58	11	24	21	24
9	49	18	23	22	28							26	46	15	17	12	17
10	57	25	28	29	30							27	53	19	17	15	18

y_0 represents initial weight of rat
y_1 gain in 1st week
y_2 gain in 2nd week
y_3 gain in 3rd week
y_4 gain in 4th week

Figure 2: Data from Box, G.E.P., "Problems in the Analysis of Growth and Wear Curves", Biometrics, Vol. 6. Dataset in short format `Rats_Box.txt`.

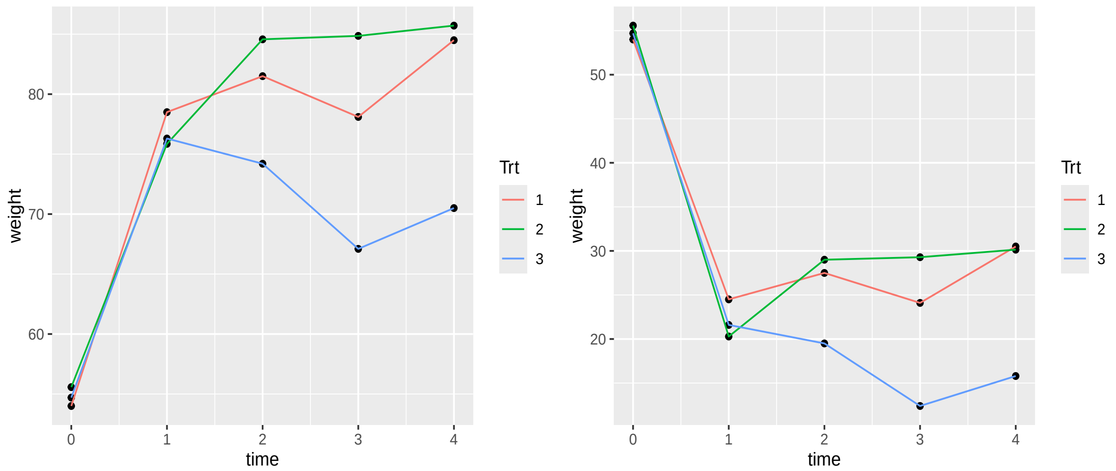

Figure 3: Observed means per week and treatment. The left plot displays the weights, and the right the initial weight and the weekly gain in weight.

Perform a statistical analysis to investigate the effectiveness of the treatments. Use a linear mixed model approach to make inferences and consider the following steps.

i) Prepare your data for analysis. Graph the response for each rat as a function of time and for the mean values for the control and two treatment groups.

ii) Do an exploratory analysis of the data, numerical and graphical.

iii) Formulate a mixed model (assuming that the observations for each rat are equally correlated). Select the best model and present it. Do model check, including diagnostic plots. Consider using time as a continuous and a discrete variable, select one option and proceed.

iv) Formulate a random effects model, now considering different correlation structures for the response. Select one model. Do model check, including diagnostic plots.

v) Include a table with the considered fitted models and some statistics.

vi) Report results on your final model and do a post hoc analysis to respond to the research question.

vii) State your main results and conclusions from your analysis. Support your main statements with quantitative results obtained from your statistical analysis/models.